Koa Shen

B.S. Candidate in Mechanical Engineering | UC Santa Barbara

koashensf@gmail.com (415) 810-8344 koashen.com San Francisco Bay Area / Santa Barbara, CA

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Zipline Mechanical Engineering Hiring Team

Re: Summer 2027 Mechanical Engineering Internship

Dear Zipline Hiring Team,

I am a mechanical engineering student at UC Santa Barbara, graduating in June 2028, and I am applying for a Summer 2027 mechanical engineering internship at Zipline. I am excited by the challenge of building autonomous aircraft that make delivery useful in the real world: the mechanisms, structures, and production details must work together reliably, not just in a model. My experience connects mechanical design and simulation with precision manufacturing, integration, and hands-on validation.

At SLAC National Accelerator Laboratory, I built a Python/Drake digital twin for a tightly packaged beamline assembly with 30 degrees of freedom and 305 bodies. Millisecond collision checks exposed at least five physical pinch points and informed mechanical redesigns. I also created 15 reusable motion-stage models, now being adopted across three assemblies, and redesigned optics mounts, alignment features, and detector strain relief. Checking load cases and producing Solid Edge designs with GD&T and PDM-controlled documentation taught me to carry design intent through analysis and into hardware that can be built and inspected.

With UCSB Gaucho Racing, I CNC-machined 33 suspension components from 4130 steel and 7075-T6 aluminum, holding critical bore and bearing fits within 0.0004- and 0.0005-inch tolerances. I worked with designers on manufacturability and inspection, and led 15 teammates through safety-system design, manufacturing, and installation for a car that passed FSAE technical inspection. In a personal actuator project, I designed a compact 15:1 cycloidal reducer and bench-validated encoder-based position correction and missed-step detection. These experiences have made me comfortable solving practical problems across design, fabrication, and test.

I would bring Zipline a hands-on engineer who cares about lightweight, manufacturable mechanisms, clear documentation, and checking hardware against its real operating constraints. I am eager to learn from a team developing aircraft systems at scale and contribute wherever careful mechanical engineering can make them more dependable and easier to build and maintain.

Thank you for your time and consideration.

Sincerely,
Koa Shen